A Subway Station Platform Crowd Simulation System

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ABSTRACT

A crowd simulation system in a subway station environment is proposed to address the problem that most of the platform passengers in the existing subway simulation systems will use fixed routes which has been planned before to get on and off the train. Along with the creation of the basic model of crowd behavior in the subway station, passengers will obtain the navigation grid and door information through the environment information generator. And, through a finite state machine and the usage of the A* algorithm which can achieve wayfinding, passengers can manage their own state in the intelligent passenger system. When selecting a screen door, passengers will use the algorithms of Topsis analysis and fitting formulas to simulate real decisions. Finally, the simulation-based passenger behavior is compared with the real situation to verify the correctness of the system.

CCS CONCEPTS

• **Computing methodologies**; • **Modeling and simulation**; • **Simulation types and techniques**;

KEYWORDS

Subway Station Crowd, Behavior simulation, A* Algorithm, Topsis analysis

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1 INTRODUCTION

The Crowd simulation techniques are widely used in evacuation simulation, urban planning, and crowd analysis. And, it studies mainly about individual human behavior and building visual models.

The Crowd simulation modeling, which has two methods, is accepted by this paper. And, this paper adopts the microscopic method to model the action of a group as a collection of individual movements. The Intelligent simulation technology of passenger's behavior is based on a crowd simulation technology and algorithms such as Agent /Multi-Agent [1]. There are three main types of passenger-based behavior modeling: meta-automata models; models based on mathematical and physical equations; models based on AGENT or some specific rules [2]. In addition, integrating emotions and personalities into individual simulations is also a popular approach to crowd simulation, which links psychological components with the individuals that make up the whole crowd and then generate behavior in the group [3]. Xu proposed an emotion-based crowd simulation model in which specific actions are determined by quantifying each individual's physical exertion and emotions of panic [4]. Based on the emotional contagion model, Mao considered the effects of stress on individual's behavior and proposed an emotion-based model of diversity behavior [5].

This paper presents a simulation of passengers'behavior in a subway station. During the whole simulation process, the behavior of each virtual passenger is calculated in real-time processing and the action of passenger model can be rendered as the same time. By investigating passengers' behavior in the subway station, this paper proposes a passenger behavior system which composed of passengers' attributes and environmental information with the support of Agent/Multi-Agent theory. The pathfinding process of passengers uses the A* pathfinding algorithm of navigation polygons to determine the least costly path between a given starting point and a target point. When the passenger is searching for the most suitable door for the ride, this paper uses two different algorithms: (1) The fitted parametric formulation, which has parameters that include the number of people waiting for the train in front of the screen door and the distance to the screen door. (2) Topsis analysis, which uses the original information to synthetically evaluate the raw data to accurately reflect the gap between the advantages and disadvantages of selecting each screen door to select the door for boarding. Finally, this paper realizes the simulation of the behavior of virtual passengers affected by multiple messages in a subway station with the help of Unity.

2 ENVIRONMENTAL INFORMATION GENERATOR

This system consists of two parts: the environmental information generator and the intelligent passenger system. The environmental information generator is responsible for generating the navigation grid, which based on the subway station model. And, the generator will passing the generated navigation grid to the intelligent

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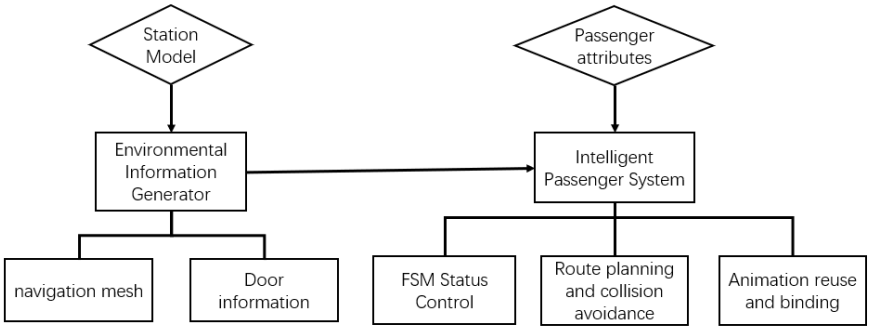


Figure 1: The general framework of passenger system

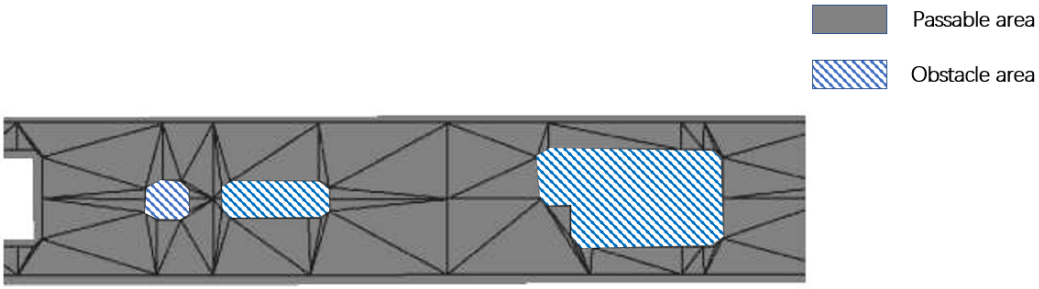


Figure 2: The generated navigation grid

passenger system for each virtual passenger, then recording each door's information. The intelligent passenger system is responsible for planning the passenger's path, determining passenger behavior, and binding the correct animation to the passenger. Also, the intelligent passenger system contains the independent attributes for each passenger, which enabling the individual randomness of passengers.

The framework of the system is shown in Figure 1

By analyzing the train station and train model data already placed in Unity, the Environment Information Generator removes impassable obstacles and areas, and then generates a navigation grid of convex polygons for the passenger intelligence system to use.

2.1 The Generation of Navigation Grid

The navigation grid, NavMesh, consists of convex polygons which used to generate walkable surface data in a virtual subway station environment, and then, it will generate a navigable mesh model. And, the model is a simple, highly intuitive planar graph. After partitioning its surface, this paper yields a triangular mesh that matches the height and shapes contours of the original geometry and forms a traversable surface. And, virtual passengers can use it to navigate and find their way around the subway station [6]. The generated navigation mesh is shown in Figure 2

2.2 The Door and Queuing Information

The uniqueness of the door storage information is ensured by creating a single instance class to save the information of the door. And, the instant class includes the location of each door, the number of passengers in line in front of the door, the visual location of passengers upon the arrival of the door and the maximum number of passengers in line in front of the door. Its class diagram is shown in Figure 3

3 INTELLIGENT PASSENGER SYSTEM

The intelligent passenger system consists of the FSM state control module, the path planning and collision avoidance module, and the animation multiplexing and binding module. The passenger attribute data is responsible for initializing each passenger's attributes, including the maximum movement speed, the collision volume of the enclosing box, and the desire for the destination to ensure that there is randomness and diversity among each passenger. The FSM finite state machine is responsible for managing the passenger's state to ensure that the passenger is in a particular state in a finite set of states at any time. When the machine has monitored that the current state can satisfy the condition for state transition, it will start changing the state and calls the corresponding action

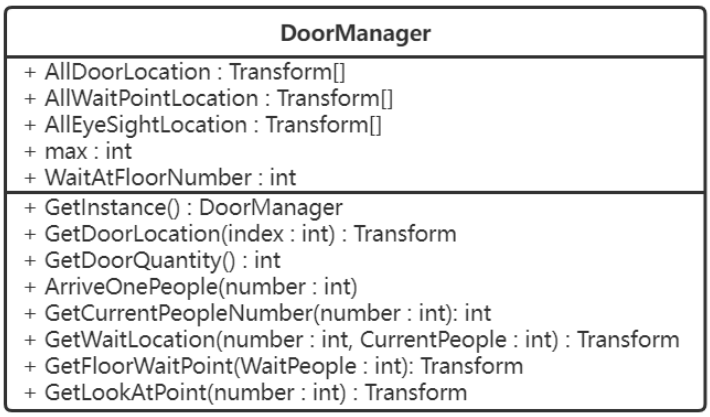


Figure 3: DoorManager class

from the behaviour library. At the same time, the machine will rescue and then match the character animation by using the Animate Controller in Unity.

3.1 FSM Finite State Machine

In this paper, an FSM finite state machine is used to control the behavioural transitions of passengers in the station. The finite state machine system is a system that exhibits different states of operation at different moments of operation, and, the states that exist in the system are non-repetitive and finite. By using the FSM finite state machine, this paper ensures that the passenger must be in one of these states during the operation of the program, and, it allows the reception of inputs to generate the corresponding state transitions.

DownStair, FindDoor, QueueUp and GetCar are the four states inherited from FSMState, which will be added to the system. When passengers are automatically generated, they will enter the DownStair state by default. When passengers go down the stairs and enter the waiting platform, they will enter the FindDoor state to find the most suitable door to get on the train. Before passengers walk to the door of the train, they will transfer to the QueueUp state to find the position they should queue up in front of the door. Finally, when the train arrives, they will transfer to the GetCar state and get on the train. Figure 4 shows the specific framework of the FSM finite state machine in the system.

3.2 Path Planning and Collision Avoidance

The A* algorithm is widely used because of its high efficiency [7]. And, the A* algorithm is used by this paper to calculate the global optimal path in a static space through a navigation grid which generated by the Environment information generator.

The A* algorithm finds the path by iteratively checking the least costly unexplored location stored in the list [8]. When a location is explored, the algorithm ends if it is the navigation target; otherwise, the algorithm will further record all the neighbouring nodes of the node for calculation. The expression of A* algorithm to calculate

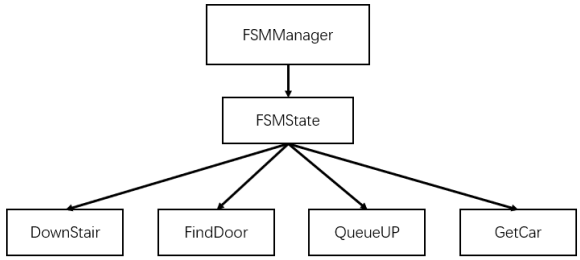


Figure 4: FSM system framework

the optimal path estimation function f(n) is shown in formula (1).

$$f(n) = g(n) + h(n) \tag{1}$$

Among them, f(n) is the integrated priority for node n. When the A* algorithm selects the next node to be traversed, the algorithm always chooses the node with the highest integrated priority (the node with the smallest value of f(n)). In which g(n) is the cost of node n from the starting point, and h(n) is the expected cost of node n from the endpoint, which is also the heuristic function of the A* algorithm [9]. A triangle is used as an example in the navigation grid, which contains the midpoints of three edges A, B, and C, as shown in Figure 5. Assuming that the previous vertex is A, the valuation function f(n) for point B is:

$$g(n) = g(n-1) + k(n) \tag{2}$$

$$h(n) = \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2} \tag{3}$$

In formula (2) and formula (3), g(n-1) is the cost from the starting point to node A, and k(n) is the Euclidean distance from point A to point B [10]. And, the A* algorithm obtains the optimal path by calculating all the nodes passed through and finding the nodes with the smallest cost to connect in order.

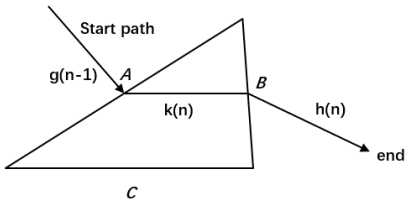


Figure 5: Diagram of A* algorithm using navigation grid

After obtaining the globally optimal path in the static environment, Rigid Body and Capsule collider are added to the program by this paper for each passenger to ensure that the passengers can move under the physical influence and achieve dynamic collision avoidance, in order to prevent the collision between passengers which result in the phenomenon of model glitch [11]. According to the general Chinese human body dimensions provided in the literature [12], this paper makes an approximation of the vertical projection of the human body as a circular projection with an area of 0.198m to pursue the simplicity of the collision body. Therefore, this paper set the radius of the passenger's capsule collider to 0.25m to ensure the rationality of the passenger's collision body.

3.3 The Design of Passengers' Behavior in Station

In the specific investigation of the subway for the behaviours of passengers on board, this paper found that the passenger behaviours has the following characteristics:

(1) Passengers' choice of boarding door is influenced by the people around them.

(2) The passenger's choice of boarding door is related to the distance from himself to the door.

(3) When passengers choose the boarding door, the door which located in the blind spot of their vision will be ignored.

(4) For the reason that the passenger's judgement of which door is optimal to board is not precise, the comprehensive calculation parameters of this paper are fuzzy estimate in large.

According to the characteristics of passengers riding in the car listed above, the behaviours of passengers when choosing the door can be simulated.

3.3.1 Door selection based on fitting formula. When entering the station, passengers will select the optimal door within their visual range to board the train. A composite parameter of the door is set for each train to quantify which door will be selected from the perspective of the passenger. The composite parameter of the door, which is S, is defined in formula (4):

$$S = \left(10 * ln\,(P + e) + \sqrt{L} * \frac{1}{M}\right) * K \tag{4}$$

In formula (4), S is the composite parameter of each door, and passengers will choose the door with the smallest door parameter to wait in line. L is the distance between passengers and that door, and P is the number of people in the queue at that door. At the beginning, P value has a significant influence on the S value, but with the increase of P value, the influence of P value on the S value decreases, which conforms to the general law of the passengers' selection of door in station. M is the number of people around the door, and K is a random value that fluctuates between 0.8 and 1.2, which in order to ensure that each passenger has a certain degree of randomness in choosing the door at the station.

3.3.2 Door selection based on Topsis analysis. TOPSIS analysis, one of the most popular MCDM/A methods [13], is a commonly used comprehensive evaluation method that makes maximum use of the information from the raw data and calculates the Euclidean distance between each evaluation solution, and, the optimal and inferior solutions can also be calculated separately to derive the ranking among the solutions [14].

This paper uses the distance between the passenger and the door as well as the number of passengers in front of the door to rate the comprehensive score of each door. Since these two metrics are minimal (the smaller, the better), they need to be normalized into extra-large metrics when in use. In addition, this paper considers that passengers will prefer the doors they can see, so it adds the indicator of whether the doors can be directly seen by passengers coming down the stairs is. And, the parameter of doors that can be directly seen is set to 1, and doors that cannot be directly seen is set to 0.3.

Converting minimal indicators to extensive indicators is shown in formula (5), where max is the maximum value among all the same indicators.

$$y = \max - y \tag{5}$$

After the normalization, it is assumed that there are n objects that need to be evaluated and m evaluation indicators normalized. Their composition of the normalization matrix is shown in formula (6).

$$\begin{matrix} y_{11} & \cdot s & y_{1m} \\ \vdots & \ddots & \vdots \\ y_{n1} & \cdot s & y_{nm} \end{matrix} \tag{6}$$

In order to eliminate the influence of different index levels on the evaluation, it is necessary to make a further standardization of the matrix which had been normalized before. The standardized matrix is denoted as Z. Each element of Z_{ij} in matrix Z is calculated as shown in formula (7). And, after the standardization, matrix Z is obtained as shown in formula (8).

$$z_{ij} = y_{ij} / \sqrt{\sum_{i=1}^{n} y_{ij}^2} \tag{7}$$

$$\begin{pmatrix} z_{11} & \cdot s & z_{1m} \\ \vdots & \ddots & \vdots \\ z_{n1} & \cdot s & z_{nm} \end{pmatrix} \tag{8}$$

Next, defines the maximum value Z+ and the minimum value Z- as shown in formula (9) and (10).

$$\begin{aligned} Z^+ &= \left(Z_1^+, Z_2^+, \ldots Z_m^+\right) = \left(\max\left\{z_{11},\ z_{21}, \ldots, z_{n1}\right\},\right. \\ &\quad \left.\max\left\{z_{12},\ z_{22}, \ldots, z_{n2}\right\}, \ldots, \max\left\{z_{1m},\ z_{2m}, \ldots, z_{nm}\right\}\right) \end{aligned} \tag{9}$$

$$\begin{aligned} Z^- &= \left(Z_1^-, Z_2^-, \ldots Z_m^-\right) = \left(\min\left\{z_{11},\ z_{21}, \ldots, z_{n1}\right\},\right. \\ &\quad \left.\min\left\{z_{12},\ z_{22}, \ldots, z_{n2}\right\}, \ldots, \min\left\{z_{1m},\ z_{2m}, \ldots, z_{nm}\right\}\right) \end{aligned} \tag{10}$$

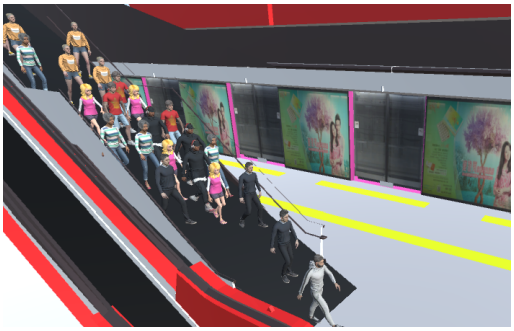


Figure 6: Passengers entering the station

Defines the distance Di^+ of the i number evaluation object from the maximum value and the distance Di^- from the minimum value shown in formula (11)and (12), ω_j are the weights of the indicators, whose sum is 1.

$$D_i^+ = \sqrt{\sum_{j=1}^{m} \omega_j \left(Z_j^+ - z_{ij}\right)^2} \tag{11}$$

$$D_i^- = \sqrt{\sum_{j=1}^{m} \omega_j \left(Z_j^- - z_{ij}\right)^2} \tag{12}$$

The final score for the i number object is derived by using formula (13).

$$S_i = \frac{D_i^-}{D_i^- + D_i^+} \tag{13}$$

After getting the composite score S for each ride door, passengers will choose the door with the highest composite score to wait in line for boarding.

When there are too many passengers in the waiting area, passengers who coming down the stairway will not continue to wait in front of the door, instead, they will choose an area closer to the door to wait. The stops chosen by passengers have a strong randomness, but they tend to choose the locations closer to the doors and less crowded for standing.

4 SIMULATION RESULTS

The intelligent passenger system is implemented based on the design of the above technology. Based on the provided model of subway station, this paper carries out the experiments, where passengers with different attributes are continuously generated at the station entrance and then enter the station as shown in Figure 6. Passengers will choose the appropriate door to wait for boarding after entering the station, as shown in Figure 7. When there are too many passengers waiting at the upper door, passengers will choose to wait for boarding at a random point in the station platform as shown in Figure 8

4.1 Realistic Results of Door-Finding Process

To verify the correctness of the simulation, a field study was conducted at the Xingye North Street Station of Chengdu Metro Line 6 at 5:00 pm. At this time, the interval between train departures was about one train per four minutes, and the five nearest doors

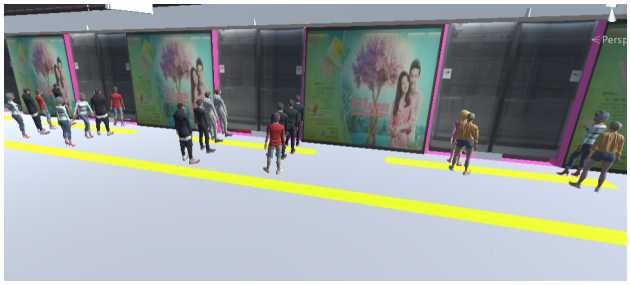


Figure 7: Passengers queuing for trains



Figure 8: Passengers waiting for trains in the waiting area

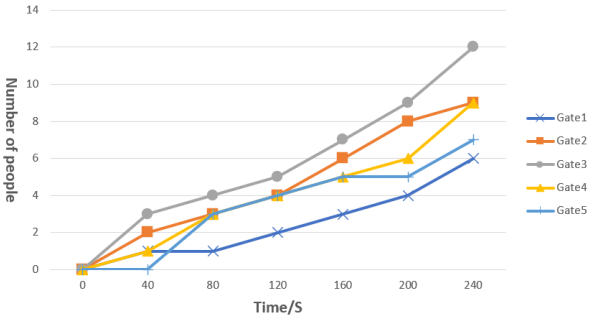


Figure 9: The number of people waiting for each screen door in the actual situation

after coming down from the platform escalator were selected for counting. The number of passengers waiting for trains at the platform doors over time is shown in Figure 9. The distance between the doors and the staircase exit are Gate3, Gate2, Gate4, Gate1 and Gate5, where Gate1 is on the left rear side of the staircase and cannot be seen directly.

4.2 Simulation Results of Door-Finding Process

In the simulation, according to the actual situation, the interval of generating passengers at the stairway is set to be about 0.7 seconds.

4.2.1 Based on fitting formula. Based on the fitting formula, the variation of the number of simulated passengers waiting for trains at the platform doors with time is shown in Figure 10

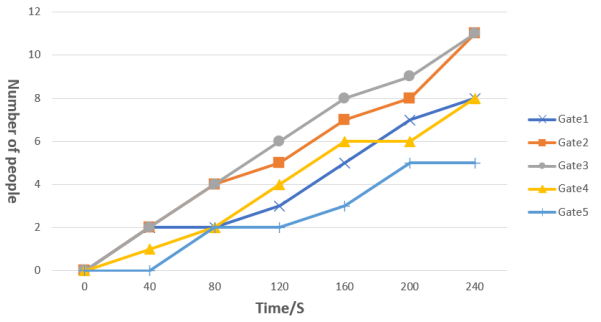


Figure 10: Number of people waiting for trains at each screen door based on the fitting formula

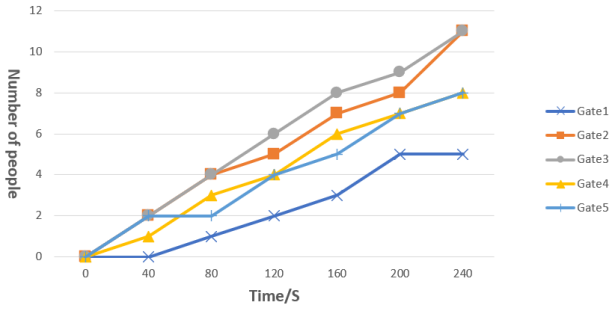


Figure 11: Weights are 0.4, 0.4, 0.2

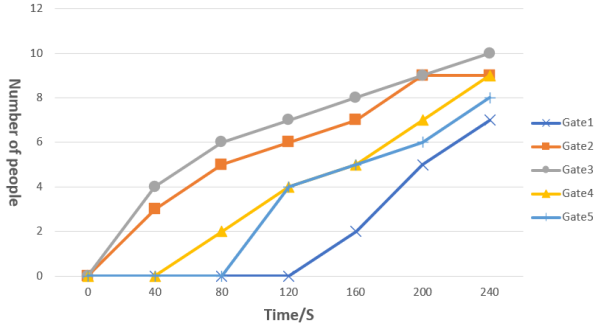


Figure 12: Weights are 0.2, 0.6, 0.2

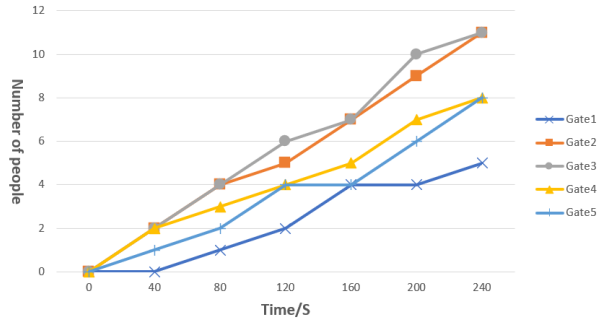


Figure 13: Weights are 0.6, 0.2, 0.2

The experimental results show that there is no delay between the passengers' choices for the doors when the simulation is based on the fitting formula. The number of people in front of each door rises slowly with the increase of simulation time, which indicates that the difference of the integrated parameter S between each door is slight and the differences between passengers' choices of each door are not obvious. However, this method does not consider the fact that passengers will prefer doors that can be seen directly and therefore will prefer Gate1, which cannot be seen directly but is closer.

4.2.2 Based on Topsis analysis. In formula (11) and (12), by adjusting the weight which use ω_j as index, this paper obtains the distance from the passenger to the door, the number of people in front of the door and whether the door can be seen directly, the influence of three parameters on the simulation of passenger behaviours are shown in Figure (11) (12) (13). In Figure 11, the distance weight is 0.4, the number of passengers is 0.4, and the weight of whether the door can be seen directly is 0.2. In Figure 12, the proportions assigned in the same order are 0.2,0.6,0,2. In Figure 13, the proportions assigned in the same order are 0.6,0.2,0,2.

The simulation results obtained when assigning weights of 0.4,0.4,0.2 are similar to the simulation based on the fitted formula, indicating that passengers have a more even choice of doors and prefer to choose the door which is more closely to them to get in the train. However, less of the passengers will choose Gate1, which is not directly visible.

With the assigned weights of 0.2,0.6,0.2, it can be seen that passengers will choose the closer door even if there are more people at the closer door and will not choose another door until there are too many people in front of it, which is the result that the distance is given less weight.

And, the same trend can be seen when the assigned weights are 0.6,0.2,0.2. However, for the reason that the weight of the number of people in front of the door is minor, so when the number of people in line in front of the door is more extensive, passengers will prefer to choose the door further away.

By comparing the above three cases with the actual situation, this paper find that the simulation results it obtained are similar to the actual ones when the parameters are set to 0.4,0.4,0.2. It means that passengers will choose the farther door to wait for the train when there are more people in front of the door, and they will not tend choose the door that cannot be seen directly even if it is closer.

5 CONCLUSION

This paper proposes an intelligent passenger simulation method based on the urban subway station scenario. In the intelligent passenger pathfinding process, this paper achieves the optimal path search by using the A* algorithm which based on the navigation grid, and, the capsule collider is also used by this paper to ensure that passengers can avoid collisions normally.

Through the results of field research in subway stations, this paper summarizes and analyze the optimal door selection strategy of

subway passengers and design a complete set of subway passenger simulation system.

In the passenger model proposed by this paper, passengers appear as individual objects, and there is no social force model such as following and moving in pairs. The inclusion and processing of such social force models will be the next step of research.

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